LIST

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LIST

A list is a versatile and mutable data structure in Python that is used to store multiple items in a single variable. Items in a list are ordered, changeable, and allow duplicate values.

Syntax:
list_name = [item1, item2, item3, ...]

Example:
list1=["apple", "banana", "cat"]

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Key Rules:

1. Square Brackets: Lists are enclosed in [ ].
2. Comma-Separated: Items in the list are separated by commas.
3. Can Be Empty: You can create an empty list using [].
4. Elements Can Be Any Data Type: A list can include integers, strings, floats, and even other lists.

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Key Characteristics of Lists:

1. Ordered: The order of elements in a list is preserved.
2. Mutable: Elements in a list can be changed after the list is created.
3. Allows Duplicates: Lists can have multiple items with the same value.
4. Heterogeneous: Lists can store elements of different data types.

LIST EXAMPLE

Empty list

```
empty_list = []
```

List of integers

```
numbers = [1, 2, 3, 4, 5]
```

List of strings

```
fruits = ["apple", "banana", "cherry"]
```

LIST EXAMPLE

Mixed data types

```
mixed = [42, "hello", 3.14, True]
```

Nested list

```
nested = [[1, 2], [3, 4]]
```

ITERATING THROUGH A LIST

Using for Loop:

```
for fruit in fruits:
    print(fruit)
```

Using Indexes:

```
for i in range(len(fruits)):
    print(fruits[i])
```

ACCESSING LIST ELEMENT

A. Indexing:

```
fruits = ["apple", "banana", "cherry"]
print(fruits[0])
```

B. Slicing:

```
numbers = [1, 2, 3, 4, 5]
print(numbers[1:4])
```

LIST OPERATION

a. Adding Elements:

```
fruits = ["apple", "banana"]
fruits.append("cherry")         # Adds at the end
fruits.insert(1, "blueberry")  # Adds at index 1
print(fruits)  # Output: ['apple', 'blueberry', 'banana', 'cherry']
```

LIST OPERATION

b. Removing Elements

```
fruits.remove("banana")  # Removes first occurrence of "banana"
fruits.pop(1)            # Removes item at index 1
fruits.clear()           # Removes all items
```

c. Modifying Elements

```
fruits[0] = "kiwi"
print(fruits) # Output: ['kiwi', 'banana']
```

QUESTIONS

a. Write a Python program to count the number of even and odd numbers in a given list.
b. Write a Python function that takes a list of numbers and returns the sum of all the elements.
c. Write a function that takes a list of integers and returns the maximum value in the list.
d. Write a Python program that takes a list of numbers and returns the square of all elements.
Input: [2, 4, 3, 7]
Output: [4, 16, 9, 49]

THANK YOU

FOR COMING